Experiment - 2

AIM

To convert the ER diagram of an Indian E-commerce Platform into a relational database schema and implement the database using MySQL with primary keys, foreign keys, unique constraints, Not Null constraints, Composite keys, ON DELETE CASCADE and ON DELETE SET NULL.

THEORY

A relational schema represents the entities of an ER Diagram in the form of tables.

Each entity is converted into a table Primary key are used to uniquely identify records, while foreign keys are used to establish relationship between tables.

A multi-valued attribute is represented using a seperate table. A weak entity is represented using a table with a composite primary key.

In this experiment, the ER diagram is converted in to relational table such as customer, address, product, seller, category, order, orderItem, Payment and Delivery.

RELATIONAL SCHEMA

The ER diagram is converted into the following relational tables:

1 - CUSTOMER
- customer_id - Primary Key
- customer_Name
- Email - Unique
- Phone

2 - RESTAURANT
- Restaurant_ID - Primary Key
- Restaurant_Name
- Location

3 - DELIVERY_AGENT
- Agent_ID - Primary Key
- Agent_Name
- phone - Unique

4 - ORDERS
- order_ID - Primary Key
- customer_ID - Foreign Key
- restaurant_ID - Foreign Key
- Agent_ID - Foreign Key
- Order_date

5 - ORDER_ITEM
- order_item_ID - Primary Key
- order_ID - Foreign Key
- Item_Name
- Quantity
- Price

SOURCE CODE

```
CREATE DATABASE
FoodDeliveryDB;
USE FoodDeliveryDB;

CREATE TABLE Customer (
customer_ID INT PRIMARY KEY,
customer_Name VARCHAR(50)
NOT NULL,
    Email VARCHAR(100) UNIQUE
NOT NULL,
    Phone VARCHAR(15)
);
CREATE TABLE Restaurant (
   Restaurant_ID INT PRIMARY KEY,
   Restaurant_Name VARCHAR(50)
NOT NULL
   Location VARCHAR(50)
NOT NULL
);
CREATE TABLE Delivery_Agent (
   Agent_ID INT PRIMARY Key,
   Agent_Name VARCHAR(50)
NOT NULL
   Phone VARCHAR(15) UNIQUE
);
CREATE TABLE Orders (
    order_ID INT PRIMARY KEY,
    customer_ID INT NOT NULL,
    Restaurant_ID INT NOT NULL,
```

```
    Agent_ID INT,
    Order_Date DATE NOT NULL,
    FOREIGN KEY (customer_ID)
    REFERENCES
Customer (customer_ID)
    ON DELETE CASCADE

    FOREIGN Key (Restaurant_ID)
    REFERENCES
Restaurant (Restaurant_ID)
    ON DELETE CASCADE,

    FOREIGN KEY (Agent_ID)
    REFERENCES
Delivery_Agent (Agent_ID)
    ON DELETE SET NULL
);
CREATE TABLE order_item(
    order_item_id INT PRIMARY KEY,
    order_id INT NOT NULL,
    item_name VARCHAR(50) NOT NULL,

    Quantity INT NOT NULL
    Price DECIMAL(10, 2) NOT NULL,

    FOREIGN KEY (order_ID)
    REFERENCES order (order_ID)
    ON DELETE CASCADE
);
```

```
INSERT INTO Customer Values.
(1, 'Shayan', shayan@gmail.com ; '8974733474'),
(2, 'Ravi', Ravi@gmail.com, '8974733471'),
(3, 'Rahul', Rahul@gmail.com, '8874733741');

INSERT INTO Restaurants VALUES
(101, 'FoodPoint', 'Chandigarh'),
(102, 'Pizza Point', 'Mohali'),
(103, 'Burger House', 'Delhi');

INSERT INTO Delivery-Agent values
(201, 'Amit', '9876598765'),
(202, 'Ravi', '9876598766');

INSERT INTO ORDERs Values
(1001, 1, 101, 201, '2026-08-19'),
(1002, 2, 102, 202, '2026-08-19'),
(1003, 3, 103, 203, '2026-08-19');

INSERT INTO ORDER-ITEM VALUES
(1, 1001, 'Pizza', 2, 250.00),
(2, 1002, 'Burger', 3, 120.00),
(3, 1003, 'Pasta', 4, 400.00);

SELECT * FROM Customer;
SELECT * FROM Restaurant;
SELECT * FROM Delivery-Agent;
SELECT * FROM Orders;
SELECT * FROM orders-Item;
```

```
INSERT INTO orders VALUES
(1004, 999, 101, 201, "2026-08-19);

INSERT INTO customer VALUES
(4, 'Karan', 'karan@gmail.com', 9876543213');

ON DELETE SET NULL
DELETE FROM DELIVERY_AGENT
WHERE Agent_ID = 201;
SELECT * FROM Orders;

DELETE FROM Customer
WHERE customer_ID = 1;

SELECT * FROM orders;
SELECT * FROM order_item;
```

OUTPUT

1- Referential Intergrity Violation

Command:

```
INSERT INTO orders
VALUES (1004, 999, 101, 201, '2026,-08-19');
```

output:

ERROR 1452 (23000): cannot add or update a child row, a foreign key constraints fails.

Reason:

Customer_ID 999 doest not exist in the customer table.

2- UNIQUE Constraint Violation

Command:

```
INSERT INTO customer
VALVE (4, 'Karan', Karan@gmail.com, '9876543213');
```

Output:

ERROR 1062 (23000) Duplicate entry 'karan@gmail.com'

Reason:

The email Karan@gmail.com already exists in the customer tables.

Conclusion

Thus, the ER diagram was successfully converted into relational tables. The required constraints were successfully applied and sample data was inserted. Referential Integrity violation UNIQUE constraint, ON DELETE SET NULL and ON DELETE CASCADE were successfully demonstrated.